

# 國立暨南國際大學資訊工程學系學士班必選修科目一覽表

## Undergraduate Program Course Catalog of Department of Computer Science and Information Engineering, National Chi Nan University

本表( 111 )學年度入學新生適用 Effective for students admitted in Fall, 2022

最低畢業學分：128 學分，全校共同課程 15 學分，通識領域課程 16 學分

系必修 49 學分，專業選修 42 學分，自由學分 6 學分。 Undergraduate requirements: a minimum of 128 required credits, including 15 credits in the School-Wide Curriculum, 16 credits of General Education Courses, 49 credits of Departmental Required Courses, 42 credits of Elective Courses, and 6 credits of School Elective Courses.

### 壹、全校共同課程 1. School-Wide Curriculum

| 一、共同課程（必修15學分） A. School-Wide Required Courses ( 15 credits ) |                                |                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|---------------------------------------------------------------|--------------------------------|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 課程代號<br>Course Code                                           | 課程名稱<br>Course Name            | 學分數<br>Credits | 備註<br>Notes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|                                                               | 英文一(上) English I(1)            | 2              | 自110學年度起，英文能力分班改為依照本校自辦英檢(OOPT)作依據，分為三級上課：前10%的同學將分配至「進階班英文」上課；中間80%的同學將分配至原班英文上課；後10%的同學將分配至「基礎班英文」上課。<br>Since the first semester of 110 Academic Year (September 2021), the competence-based class grouping policy of Freshmen English (Course Title: English 1) will be based on registered student's OOPT scores; Students will be accordingly divided into three different levels: Advanced English (top 10%), General English (middle 80%) and Basic English (bottom 10%).                 |
|                                                               | 英文一(下) English I(2)            | 2              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                                                               | 英文二 English II                 | 2              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                                                               | 國文(上) Chinese(1)               | 2              | 僑外生得以修習「僑外生華語文」抵免國文。(Overseas students take Chinese for Overseas Students)                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                                                               | 國文(下) Chinese(2)               | 2              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                                                               | 社會服務學習(上) Service Learning(1)  | 1              | 依所屬學系分班上課，無特殊原因不得跨班選課。(Students are divided into groups by their own department, without any proper reasons they cannot change the decision.)                                                                                                                                                                                                                                                                                                                                                     |
|                                                               | 社會服務學習(下) Service Learning(2)  | 1              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                                                               | 大一體育(上) Physical Education (1) | 0.5            | 上學期依所屬學系分班上課，無特殊原因不得跨班選課，惟未達40人之學系、院學士班班級，則採合併班開課；下學期則依運動項目分班，自由選課。<br>(In the first semester of freshman, students are arranged to take Physical Education (1) by their original department. Without any proper reasons, change will be prohibited. If a class of the department or the interdisciplinary program that is less than 40 students, the combined class will be adopted. In the second semester of freshman, students are able to take Physical Education (2) by their interesting.) |
|                                                               | 大一體育(下) Physical Education (2) | 0.5            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                                                               | 特色運動 (Featured Sports)         | 2              | 大二開始，每學期限修一項，畢業前須完成至少2項特色運動 (Students are limited take one course each semester since Sophomore, At least 2 featured sports courses are required before graduation.)                                                                                                                                                                                                                                                                                                                              |
| 通識講座 General Education lecture program                        |                                | 0              | 畢業前至少應出席6場。(Students are required to attend at least 6 lectures before graduation.)                                                                                                                                                                                                                                                                                                                                                                                                               |
| 二、共同選修課程（各系依共同選修加列課程） B. School-Wide Elective Courses         |                                |                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

|  |                                 |  |                                                                                                                                                                                             |
|--|---------------------------------|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | 第二外語 (Second Foreign Language)  |  | 含日文、韓文、法文、德文及西班牙文等。(Including Japanese, Korean, French, German and Spanish, etc.)                                                                                                           |
|  | 東南亞語言(Southeast Asia Languages) |  | 含泰語、越南語、緬甸語、印尼語及馬來語，可抵特色通識「東南亞」次領域。(Including Thai, Vietnamese, Burmese, Indonesian and Malay. These courses can be credited course credits in Southeast Asia of Topical General Education) |
|  | 進階語文(Advanced English)          |  |                                                                                                                                                                                             |
|  | 其他選修(Other Elective Course)     |  |                                                                                                                                                                                             |

## 貳、通識領域課程 (至少16學分) 2. General Education Courses (16 credits)

| 主領域<br>Main Field                   | 次領域<br>Sub-Field                         | 學分數<br>Credits | 備<br>Notes                                                                                                                            | 註 |
|-------------------------------------|------------------------------------------|----------------|---------------------------------------------------------------------------------------------------------------------------------------|---|
| 人文領域<br>Humanities                  | 文學與藝術(Literature and Art)                | 4              | 每次領域各1門，共至少修習4學分。(Students in the two sub-fields should attend at least one course, and attend a total of at least 4 credits)         |   |
|                                     | 歷史哲學與文化(History, Philosophy and Culture) |                |                                                                                                                                       |   |
| 社會領域<br>Society                     | 法政與教育(Law, Politics and Education)       | 4              | 每次領域各1門，共至少修習4學分。(Students in the two sub-fields should attend at least one course, and attend a total of at least 4 credits)         |   |
|                                     | 社經與管理(Society, Economy and Management)   |                |                                                                                                                                       |   |
| 自然領域<br>Nature                      | 工程與科技(Engineering and Technology)        | 4              | 每次領域各1門，共至少修習4學分。(Students in the two sub-fields should attend at least one course, and attend a total of at least 4 credits)         |   |
|                                     | 生命與科學(Life and Science)                  |                |                                                                                                                                       |   |
| 特色通識領域<br>Topical General Education | 東南亞(Southeast Asia)                      | 4              | 自由選擇其中兩個次領域各修習一門，共至少修習4學分。(Students should select two sub-fields courses in the main field, and attend a total of at least 4 credits) |   |
|                                     | 綠概念(Green Concept)                       |                |                                                                                                                                       |   |
|                                     | 在地實踐(Social Practice)                    |                |                                                                                                                                       |   |

## 參、系必修 (至少49學分) 3. Departmental Required Courses (49 credits)

| 課程代號<br>Course Code | 課程名稱(中文及英文)<br>Course Name (Chinese and English)          | 學分數<br>Credits | 修別<br>Course type<br>(required/elective) | 開課年級<br>Course availability<br>(freshman/sophomore/junior/senior) | 備<br>Notes  | 註 |
|---------------------|-----------------------------------------------------------|----------------|------------------------------------------|-------------------------------------------------------------------|-------------|---|
| 210036              | 計算機概論<br>Introduction to Computer Science                 | 3              | 必                                        | 一上                                                                |             |   |
| 210003              | 微積分(上)<br>Calculus(1)                                     | 3              | 必                                        | 一上                                                                |             |   |
| 210013              | 離散數學<br>Discrete Mathematics                              | 3              | 必                                        | 一上                                                                |             |   |
| 210008              | 微積分(下)<br>Calculus(2)                                     | 3              | 必                                        | 一下                                                                |             |   |
| 210099              | 程式設計<br>Programming                                       | 3              | 必                                        | 一下                                                                | 擋修PHP網頁程式設計 |   |
| 210151              | 邏輯設計與實驗(一)<br>Logic Design and Laboratory Experiments (1) | 3              | 必                                        | 一下                                                                |             |   |
| 210111              | 機率<br>Probability                                         | 3              | 必                                        | 一下                                                                |             |   |
| 210016              | 資料結構與演算法(一)<br>Data Structures and Algorithms (I)         | 3              | 必                                        | 二上                                                                |             |   |
| 210152              | 邏輯設計與實驗(二)<br>Logic Design and Laboratory Experiments (2) | 3              | 必                                        | 二上                                                                |             |   |

|        |                                                    |   |   |    |  |
|--------|----------------------------------------------------|---|---|----|--|
| 210022 | 資料結構與演算法（二）<br>Data Structures and Algorithms (II) | 3 | 必 | 二下 |  |
| 210026 | 線性代數<br>Linear Algebra                             | 3 | 必 | 二下 |  |
| 210025 | 計算機組織與結構<br>Computer Organization and Architecture | 3 | 必 | 三上 |  |
| 210093 | 系統程式<br>System Programming                         | 3 | 必 | 三上 |  |
| 210121 | 專題（一）<br>Project Study (I)                         | 2 | 必 | 三上 |  |
| 210122 | 專題（二）<br>Project Study (II)                        | 2 | 必 | 三下 |  |
| 210102 | 作業系統<br>Operating Systems                          | 3 | 必 | 三下 |  |
| 210038 | 微算機實驗<br>Microcomputer Systems Lab                 | 3 | 必 | 三下 |  |

肆、專業選修（至少42學分） 4. Elective Courses ( 42 credits )

|        |                                                        |   |   |         |  |
|--------|--------------------------------------------------------|---|---|---------|--|
| 210015 | 數位電子學<br>Digital Electronics                           | 3 | 選 | 1,2,3,4 |  |
| 130031 | 軟體工程<br>Software Engineering                           | 3 | 選 | 3,4,g,p |  |
| 210021 | 組合數學<br>Combinatorial Mathematics                      | 3 | 選 | 1,2,3,4 |  |
| 210123 | 自動機與形式語言<br>Automata and Formal Language               | 3 | 選 | 1,2,3,4 |  |
| 210124 | 數位系統設計<br>Digital System Design                        | 3 | 選 | 1,2,3,4 |  |
| 210125 | 數值方法<br>Numerical Methods                              | 3 | 選 | 1,2,3,4 |  |
| 210048 | 高等演算法（一）<br>Advanced Algorithms (I)                    | 3 | 選 | 3,4,g,p |  |
| 210126 | 電腦視覺與人機介面<br>Computer Vision and Man-machine Interface | 3 | 選 | 1,2,3,4 |  |
| 210050 | 圖形理論及應用<br>Graph Theory with Applications              | 3 | 選 | 3,4,g,p |  |
| 210052 | 數位影像處理<br>Digital Image Processing                     | 3 | 選 | 3,4,g,p |  |
| 210127 | 資料壓縮<br>Data Compression                               | 3 | 選 | 3,4,g   |  |
| 210128 | 電腦圖學<br>Computer Graphics                              | 3 | 選 | 3,4,g   |  |
| 210129 | Web技術<br>WEB Technologies                              | 3 | 選 | 1,2,3,4 |  |
| 210060 | 高等演算法（二）<br>Advanced Algorithms (II)                   | 3 | 選 | 3,4,g,p |  |
| 210082 | 虛擬實境<br>Virtual Reality                                | 3 | 選 | 3,4,g,p |  |
| 210085 | 多媒體通訊<br>Multimedia and Communication                  | 3 | 選 | 3,4,g,p |  |
| 210130 | 具體數學<br>Concrete Mathematics                           | 3 | 選 | 1,2,3,4 |  |

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| 210131 | UNIX程式設計<br>UNIX Programming                                                                          | 3 | 選 | 1,2,3,4 |               |
| 210132 | 程式語言<br>Programming Languages                                                                         | 3 | 選 | 1,2,3,4 |               |
| 210096 | 高等計算機結構<br>Advanced Computer Architecture                                                             | 3 | 選 | 3,4,g,p |               |
| 210103 | 生物計算<br>Computational Biology                                                                         | 3 | 選 | 3,4,g,p |               |
| 210133 | 隨機程序<br>Introduction to Fundamental Stochastic Processes                                              | 3 | 選 | 1,2,3,4 |               |
| 210107 | 組合語言<br>Assembly Language                                                                             | 3 | 選 | 3,4,g,p |               |
| 210134 | LISP程式設計<br>LISP Programming                                                                          | 3 | 選 | 1,2,3,4 |               |
| 210135 | UNIX使用入門<br>Introduction to Unix Environment                                                          | 3 | 選 | 1,2,3,4 |               |
| 215021 | 自然語言處理<br>Natural Language Processing (NLP)                                                           | 3 | 選 | 3,4,g,p |               |
| 210136 | 電腦網路通信協定<br>Computer Network Protocols                                                                | 3 | 選 | 1,2,3,4 |               |
| 215038 | 錯誤更正碼<br>Error Correcting Code                                                                        | 3 | 選 | 3,4,g,p |               |
| 215047 | 資訊檢索<br>Information Retrieval                                                                         | 3 | 選 | 3,4,g,p |               |
| 215049 | 史料與史學<br>Historical Sources and Historiography                                                        | 3 | 選 | 3,4,g,p |               |
| 219066 | 嵌入式系統實作<br>Embedded System Implementation                                                             | 3 | 選 | 3,4,g,p |               |
| 219068 | 非擬真成像技術<br>Non Photo-realistic Rendering                                                              | 3 | 選 | 3,4,g,p | 擋修電腦圖學及高等電腦圖學 |
| 210137 | 科技英文(一)<br>Technical English(I)                                                                       | 3 | 選 | 1,2,3,4 |               |
| 219091 | 智慧型演化計算<br>Intelligent Evolutionary Computation                                                       | 3 | 選 | 3,4,g,p |               |
| 210140 | 智慧型行動裝置軟體設計<br>Intelligent Mobile Device Programming                                                  | 3 | 選 | 1,2,3,4 |               |
| 219093 | 智慧型系統實務(一)<br>Intelligent System Practice (1)                                                         | 2 | 選 | 3,4,g,p |               |
| 219096 | Linux系統管理<br>Linux System Administration                                                              | 3 | 選 | 3,4,g,p |               |
| 219097 | 進階Linux系統管理<br>Advanced Linux System Administration                                                   | 3 | 選 | 3,4,g,p |               |
| 219101 | 語言學習與科技<br>Language Learning and Technology                                                           | 3 | 選 | 3,4,g,p |               |
| 219103 | 機器學習<br>Machine Learning                                                                              | 3 | 選 | 3,4,g,p |               |
| 219106 | 適地性文創應用與服務系統<br>Location-based Cultural & Creative Application and Service Systems                    | 3 | 選 | 3,4,g,p |               |
| 219107 | 智慧型文化導向行動導覽服務創意平台<br>技術<br>Intelligent Web Mobile Platform for Culture-Oriented and Creative Services | 3 | 選 | 3,4,g,p |               |

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| 219108 | 行動式3D電腦繪圖<br>Mobile 3D Computer Graphics                                                                              | 3 | 選 | 3,4,g,p | 擋修電腦圖學及高等電腦圖學 |
| 219110 | 地方文獻編纂與習作<br>Local Bibliography Compilation and Practice                                                              | 3 | 選 | 3,4,g,p |               |
| 210141 | 3D建模與互動多媒體應用<br>3D Modeling and Interactive Multimedia Applications                                                   | 3 | 選 | 1,2,3,4 |               |
| 219113 | 網際網路電話加值服務<br>IP Telephony Service Creation Technology                                                                | 3 | 選 | 3,4,g,p |               |
| 210142 | 數值分析<br>Numerical Analysis                                                                                            | 3 | 選 | 1,2,3,4 |               |
| 210143 | Web服務和應用<br>Web Services and Applications                                                                             | 3 | 選 | 1,2,3,4 |               |
| 210144 | 通訊導論<br>Introduction to Communications                                                                                | 3 | 選 | 3,4,g,p |               |
| 210119 | 網際網路程式設計<br>WWW Programming Design                                                                                    | 3 | 選 | 3,4,g,p |               |
| 219076 | 智慧型系統實務（三）<br>Intelligent System Practice (3)                                                                         | 2 | 選 | 3,4,g,p |               |
| 219099 | 雲端系統設計<br>Design of Cloud Computing System                                                                            | 3 | 選 | 3,4,g,p |               |
| 219115 | 音樂,大腦,與科技<br>Music,Brain,Science and Technology                                                                       | 3 | 選 | 3,4,g,p |               |
| 219116 | 英文寫作, 口說, 簡報 (一)<br>On English Writing, Speaking andPresentation (1)                                                  | 3 | 選 | 3,4,g,p |               |
| 219117 | 英文寫作, 口說, 簡報 (二)<br>On English Writing, Speaking andPresentation (2)                                                  | 3 | 選 | 3,4,g,p |               |
| 210146 | 基礎程式設計<br>Computer Fundamentals & Programming                                                                         | 1 | 選 | 1,2,3,4 |               |
| 219121 | 尖端資訊科技國際課程<br>International Studies on Leading Information Technologies                                               | 1 | 選 | 3,4,g,p |               |
| 219118 | 史學研究與資訊科技跨領域合作實務<br>When History Meets Computer Science: Theory and Practice                                          | 3 | 選 | 3,4,g,p |               |
| 219130 | 安卓程式設計<br>Android Programming                                                                                         | 3 | 選 | 3,4,g,p |               |
| 219122 | QNAP應用程式設計<br>Linux Programming for QNAP                                                                              | 3 | 選 | 3,4,g,p |               |
| 219123 | 開源電子原型平台軟硬體整合設計與實作<br>software / hardware co-design & implementation for open-source electronics prototyping platform | 3 | 選 | 3,4,g,p |               |
| 210147 | 基礎程式設計 (二)<br>C++ Programming Fundamentals                                                                            | 1 | 選 | 1,2,3,4 |               |
| 219128 | 數位音樂學習與創作<br>Digital Music Learning and Creation                                                                      | 3 | 選 | 3,4,g,p |               |

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| 219133 | 國際IT科技發展趨勢<br>Global Trend of Information Technology Development              | 2 | 選 | 3,4,g,p |               |
| 219134 | 尖端資訊科技國際課程（二）<br>International Studies on Leading Information Technologies(2) | 2 | 選 | 3,4,g,p |               |
| 210148 | 專業實習<br>Specialized Internship                                                | 3 | 選 | 4       |               |
| 210149 | 企業實習<br>Enterprise Internship                                                 | 6 | 選 | 4       |               |
| 219135 | 交響樂團與數位應用<br>Orchestra and Its Digital Applications                           | 2 | 選 | 3,4,g,p |               |
| 219136 | 音樂科技專題策展<br>Capstone Project on Music Technology                              | 2 | 選 | 3,4,g,p |               |
| 219137 | 數位錄音錄影工程<br>Digital Audio/Video Recording and Engineering                     | 2 | 選 | 3,4,g,p |               |
| 219138 | 音樂, 大腦, 與科技<br>Music, Brain, Science and Technology                           | 2 | 選 | 3,4,g,p |               |
| 219139 | 數位音樂學習與創作<br>Digital Music Learning and Creation                              | 2 | 選 | 3,4,g,p |               |
| 219140 | PHP 網頁程式設計<br>PHP Web Programming                                             | 3 | 選 | 3,4,g,p | 需先修210099程式設計 |
| 219141 | 計算智能<br>Computational Intelligence                                            | 3 | 選 | 3,4,g,p |               |
| 219142 | 大數據系統實務<br>Big Data System and Practice                                       | 3 | 選 | 3,4,g,p |               |
| 219143 | 深度學習<br>Deep Learning                                                         | 3 | 選 | 3,4,g,p |               |
| 210009 | 電子電路<br>Electrical Circuits                                                   | 3 | 選 | 1,2,3,4 |               |
| 210086 | 工程數學<br>Engineering Mathematics                                               | 3 | 選 | 1,2,3,4 |               |
| 210115 | 邏輯設計<br>Logic Design                                                          | 3 | 選 | 1,2,3,4 |               |
| 210040 | 編譯器<br>Compilers                                                              | 3 | 選 | 3,4,g,p |               |
| 210076 | 資料庫系統<br>Database System                                                      | 3 | 選 | 3,4,g,p |               |
| 210145 | 計算機網路<br>Computer Networks                                                    | 3 | 選 | 1,2,3,4 |               |
| 210023 | 數位電路實驗<br>Digital Circuits Lab                                                | 1 | 選 | 1,2,3,4 |               |
| 210029 | 微計算機系統<br>Microcomputer Systems                                               | 3 | 選 | 1,2,3,4 |               |
| 210150 | 基本電學<br>Basic Electricity                                                     | 3 | 選 | 1,2,3,4 |               |
| 210153 | MFC視窗程式設計<br>Windows MFC Programming                                          | 3 | 選 | 1,2,3,4 | 需先修程式設計       |
| 210154 | 個人電腦概論<br>Information Technology Essentials                                   | 3 | 選 | 1,2,3,4 | 需先修計算機概論      |

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| 219146 | 程式練習及訓練與競賽課程<br>Programming Practice, Training, and Contest Course                                         | 3 | 選 | 3,4,g,p |                  |
| 210155 | Web程式設計<br>Web Programming                                                                                 | 3 | 選 | 3,4,g,p |                  |
| 210156 | JavaScript程式設計<br>JavaScript Programming                                                                   | 3 | 選 | 1,2,3,4 | 需先修程式設計          |
| 219148 | SQL程式設計<br>SQL Programming                                                                                 | 3 | 選 | 3,4,g,p | 大學部必須先修通過資料庫系統   |
| 219149 | 程式練習及訓練與競賽（二）<br>Programming Practice, Training, and Contest Course (II)                                   | 3 | 選 | 3,4,g,p |                  |
| 219150 | 前端程式設計<br>Frontend Programming                                                                             | 3 | 選 | 3,4,g,p | 大學部必須先修通過Web程式設計 |
| 219151 | IT科技日文<br>IT/Technical Japanese                                                                            | 3 | 選 | 3,4,g,p | 需先修日文一(上)、日文一(下) |
| 219152 | Python網頁擷取程式設計<br>Web Scraping with Python                                                                 | 3 | 選 | 3,4,g,p | 需先修計算機概論         |
| 215050 | MXNET之深度學習實驗<br>MXNET-based Deep Learning Lab                                                              | 3 | 選 | 2,3,4,g |                  |
| 219155 | Python機器學習實務<br>Machine Learning Practice with Python                                                      | 3 | 選 | 3,4,g,p |                  |
| 219156 | iOS App 程式開發<br>iOS App Programming                                                                        | 3 | 選 | 3,4,g,p |                  |
| 215051 | C 語言教學探討<br>Discussion and Inquiry of Learning C                                                           | 3 | 選 | 3,4,g   |                  |
| 219157 | FPGA實作深度學習方法<br>FPGA Implementation of Deep Learning Methods                                               | 3 | 選 | 3,4,g,p | 需先修邏輯設計或邏輯設計與實驗  |
| 215052 | VR/AR應用實作<br>VR/AR - Implementation and Applications                                                       | 3 | 選 | 3,4,g,p | 需先修電腦圖學          |
| 219159 | 情境模擬控制系統之軟硬體整合應用實務<br>Hardware / Software Integration & Practice for Situational Simulation Control System | 3 | 選 | 3,4,g,p |                  |
| 219160 | 智慧型積木機器人程式設計實務 EV3<br>Intelligent Lego Robot Programming                                                   | 3 | 選 | 3,4,g,p |                  |
| 219161 | 音樂程式設計<br>Music Programming                                                                                | 3 | 選 | 3,4,g,p |                  |
| 210106 | 電腦與資訊安全<br>Computer and Information Security                                                               | 3 | 選 | 3,4,g,p |                  |
| 210157 | 企業實習240(一)<br>Enterprise Internship 240(一)                                                                 | 3 | 選 | 4       |                  |
| 210158 | 企業實習240(二)<br>Enterprise Internship 240(二)                                                                 | 3 | 選 | 4       |                  |
| 210159 | 企業實習(二)<br>Corporate Internship (2)                                                                        | 6 | 選 | 4       |                  |
| 219169 | 資訊科學特論<br>Special Topics in Computer Science                                                               | 3 | 選 | 3,4,g,p |                  |
| 219168 | 物聯網晶片開發實務<br>Practices of Internet of Things Chip Development                                              | 3 | 選 | 3,4,g,p |                  |

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|----------------------------------|--------------------------------------------------------|---|---|-----------|--|
| 215058                           | 互動音樂程式設計與創作<br>Interactive Music Program and Creation  | 1 | 選 | 1,2,3,4,g |  |
| 210161                           | Linux系統管理<br>Linux System Administratio                | 3 | 選 | 1,2,3,4   |  |
| 219171                           | 深度學習實驗<br>Deep Learning Lab                            | 3 | 選 | 3,4,g,p   |  |
| 215056                           | 程式設計與運算思維<br>Programming and Computational Thinking    | 3 | 選 | 1,2,3,4,g |  |
| 215057                           | 程式設計與美學創作<br>Creative Coding and Aesthetic Programming | 3 | 選 | 1,2,3,4,g |  |
| 專業選修42學分：開課年級依各課程決定，修課建議請參考課程地圖。 |                                                        |   |   |           |  |





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